ESP32 + WS2815 WLED

Bill of Materials & Quotation

Document Date: March 31, 2026
Revision: 1.0
Project: LED Controller Dual Channel
Scope: 2× ESP32 + 2× WS2815 Strip

1. Project Summary

Parameter	Value	Parameter	Value
Microcontrollers	2× ESP32-WROOM-32D	**Firmware**	WLED (open-source)
LED Strips	2× WS2815 5 m / 60 LED/m / 12 V	**Data GPIO**	GPIO4 per ESP32
Total LEDs	600 (300 per strip)	**Level Shifter**	SN74AHCT125N (3.3 V→5 V)
Strip Voltage	**12 V DC (direct from PSU)**	**Power Topology**	12 V direct to strip; small buck for ESP32
Peak Strip Current	≈7.5 A per strip (full white)	**PSU per System**	2× Mean Well LRS-150-12 (one per location)
Power per System	≈120 W (2 independent systems)	**Safety Margin**	≥30% on all power ratings

2. Power Calculation Summary

*Each system is fully independent at a separate location. Do **not** combine the two systems — values below are **per system**.*

Load	Voltage	Current (worst case)	Power	With 30% Margin
WS2815 Strip (300 LEDs × 20 mA, 18 W/m)	12 V	7.50 A	90.0 W	9.75 A / 117 W
ESP32 + Level Shifter (via small 12 V→5 V buck)	5 V	0.26 A	1.30 W	0.34 A / 1.7 W
ESP32 supply at 12 V input (buck efficiency 85%)	12 V	0.12 A	1.50 W	0.16 A / 1.9 W
TOTAL per system at 12 V	**12 V**	**7.62 A**	**91.5 W**	**9.9 A → 10 A fuse**
PSU required per system	**12 V**	**≤12.5 A**	**≤120 W**	**LRS-150-12 (12.5 A) ✓**

Wire Drop Check (WS2815 12 V, AWG12, 5 m)

At 7.5 A through AWG12 over 5 m: round-trip drop = 7.5 A × 52 mΩ = **0.39 V**. Strip entry sees **11.61 V** — above WS2815 minimum of **11.5 V**.

Power injection at both ends is RECOMMENDED. Internal strip trace resistance (≈30 mΩ/m) adds further drop at high brightness. With injection at both ends, far-end voltage remains ≥11.6 V.

3. Bill of Materials (BOM)

#	Component	Model / Part#	Qty	Specifications	Unit (BRL)	Total (BRL)	Product Link
1	ESP32 DevKit	ESP-WROOM-32 NODEMCU 38P	2	3.3 V I/O, 240 MHz, WiFi/BT, 5 V VIN	R$77,39	R$154,78	ESP-WROOM-32 NODEMCU 38P
2	WS2815 LED Strip	5 m 12 V IP20 60 LED/m 300 LEDs	2	12 V, 25 mA/LED, 300 LEDs, GRB order	R$167,20	R$334,40	WS2815 LED strip 5m 12V
3	Buck Conv. (ESP32)	XL4015 12 V→5 V 5 A module	2	Input 8–36 V, output 5 V / 5 A adj.	R$13,76	R$27,52	XL4015 Step-Down 5A
4	Level Shifter IC	74HC125 (DIP-14) subst. AHCT125N	2	Quad buffer; DIP-14; Vih ≥ 3.15 V@5 V (HC, not AHCT)	R$1,41	R$2,82	74HC125 Buffer DIP-14
5	Capacitor 470 μF/25 V	Generic 470uF/25V 105°C	2	Electrolytic, radial, **25 V rated** (12 V rail)	R$0,55	R$1,10	Cap. Elet. 470uF/25V
6	Capacitor 470 μF/25 V	Generic 470uF/25V 105°C	2	Electrolytic, radial, 105 °C rated	R$0,55	R$1,10	Cap. Elet. 470uF/25V
7	Resistor 330 Ω	CR25 1/4W 330R ±5% carbon film	2	330 Ω, 0.25 W; DIN signal protection	R$0,07	R$0,14	Resistor CR25 1/4W 330R
8	Blade Fuse + Holder	Blade Mini **10 A** + Fuse Holder	2	**10 A** / 32 V DC; per-system protection	R$6,63	R$13,26	Blade Fuse 10A Fuse Holder Mini
9	Terminal Block 2-pin	2EDGK Terminal 5.08 mm 2-Way	8	Screw terminal, 5.08 mm pitch	R$1,87	R$14,96	Terminal 2EDGK 5.08mm 2W
10	Power Wire 4 mm^2	Flex. Cable 4 mm^2 750 V Black	4 m	20 A, flexible; 12 V power runs	R$5,06/m	R$20,24	Flex. Cable 4mm^2 750V/m
11	Signal Wire AWG22	Thin Wire 0.32 mm^2 300 V 70°C	2 m	Low capacitance; DIN data lines	R$1,42/m	R$2,84	Thin Wire 0.32mm^2 300V/m
12	Heatsink	Heatsink DN812 27×16×30 mm	2	Aluminium extruded; for XL4015 IC	R$3,69	R$7,38	Heatsink DN812 27x16x30
13	Connector 2-pin Pairs	2-Pin Connector Male+Female	6	Pairs w/ crimp terminals	R$5,91	R$35,46	2-Pin Square Connector
14	Ferrule Crimps 4 mm^2	Ferrule Kit 0.5–6 mm 400 pcs	1 kit	Wire-end treatment for screw terminals	R$29,73	R$29,73	Ferrule Kit 400 pcs
					Components subtotal:	**R$645,85**	
					Shipping (contact Proesi):	—	
					TOTAL ESTIMATED PROJECT COST:	**R$645,85**	

Proesi (proesi.com.br) prices verified on March 31, 2026. Items 2 (WS2815) and 3 (XL4015) are not stocked by Proesi — prices sourced from Mercado Livre; verify current pricing before ordering. Product links open Proesi or Mercado Livre pages — confirm stock before purchasing. Proesi phone: (47) 3080-5529.

4. Engineering Notes & Recommendations

Topic	Detail
Level Shifter (REQUIRED)	ESP32 GPIO high = 3.3 V; WS2815 $V_{\mathrm{ih,min}}$ = 3.5 V. Margin: **−200 mV** (violation). The SN74AHCT125N accepts 3.3 V input ($V_{\mathrm{ih,min}}$ = 2.0 V) and drives output to full 5 V. Omitting this causes intermittent "zombie pixel" failures that vary by LED batch, temperature, and cable length.
Power Injection (RECOMMENDED)	External wire drop (AWG12, 5 m, 7.5 A) = **0.39 V**. Strip entry sees **11.61 V** — above WS2815 minimum of **11.5 V**. However, internal strip trace resistance (≈30 mΩ/m) adds further drop at high brightness. Inject 12 V/GND at **both ends** of each strip.
Wire Gauge	Use **AWG12** for the 12 V supply bus and strip runs (7.5 A at 12 V; 0.39 V drop over 5 m — within WS2815 spec). Use AWG22 for DIN data lines.
Buck Derating	WS2815 strips connect **directly at 12 V** — no strip buck needed. ESP32: one XL4015 5 A per unit, adequate for 0.5 A max draw. Mount heatsinks; derate to 70% for continuous operation.
WLED Setup	Navigate to `Config` → `LED Preferences`: set **LED Type = WS2815** (not WS2812B) and **Maximum Current = 6000 mA** per strip (protects a 10 A PSU; raise to 7500 mA if using LRS-150-12). Synchronise both ESP32 units via WLED multicast UDP (port 21324, same subnet).
Fusing	Install a **10 A blade fuse** on the 12 V feed of each system. Strip worst-case current is 7.5 A; 10 A fuse provides protection with margin. Do not use 25 A fuses — they will not protect AWG12 wiring at strip-level faults.

5. Pre-Assembly Safety Checklist

Verify every item before energising the system for the first time (per system).

☐	**10 A** fuse installed on each 12 V feed	☐	WLED LED type set to **WS2815** (not WS2812B)
☐	Wire gauge verified: AWG12 min. for all **12 V** runs	☐	WLED max current set to **6000 mA** per strip
☐	PSU output verified at 12.0 V before connecting strip	☐	SN74AHCT125N level shifter on each DIN line
☐	Buck output trimmed to 5.00 V before connecting ESP32	☐	330 Ω series resistor on each DIN line
☐	Common GND: PSU, buck, ESP32 and strip (star topology)	☐	470 μF/25 V cap at each strip 12 V entry point
☐	Power injection at **both ends** of each 5 m strip	☐	Heatsink installed on XL4015 with thermal compound
☐	All stranded wire ends crimped with ferrules	☐	WLED firmware flashed and AP connection verified

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PSU Sizing Note — 10 A vs 12.5 A

Q: Is a 10 A / 220 V → 12 V PSU sufficient?
Yes, for typical use. The 7.5 A figure is the absolute worst case: all 300 LEDs at *full white* (R + G + B at 100%) simultaneously. In practice, colorful effects draw **2–4 A** per strip.

Scenario	Strip current	10 A PSU load
Full white, 100% (worst case)	7.5 A	75%
Colorful effects, 80%	≈3–5 A	30–50%
Single colour, 50%	≈1.5 A	15%

A 10 A PSU is acceptable provided: (1) WLED maximum current is set to **7500 mA**; (2) the PSU is a quality unit (Mean Well or equivalent), not a no-brand transformer.
The LRS-150-12 (12.5 A) in the BOM provides a better safety margin and is recommended. Both systems are independent — simultaneous use has no effect on per-system sizing.

Document generated March 31, 2026. Revision 2.0 — corrected for WS2815 @ 12 V, independent systems. Power calculations validated via analytical SPICE model (AWG12, 5 m, 7.5 A). This document covers 2 independent systems at separate locations (2× ESP32 + 2× WS2815 strips, 2× separate PSUs). All designs are provided for reference only — verify all specifications before assembly.